Mapping (Version 1.0)
`Crypto_AEAD_DecryptVerify(K, C, A, N) := {boolean, byte[lengthInBytes(P)]} AES-CCM-DecryptVerify(K := K, C := C, A := A, N := N, Tlen := CRYPTO_AEAD_MIC_LENGTH_BITS)`
`AES-CCM-DecryptVerify()` SHALL be the function described in Section 6.2 of NIST 800-38C with the counter generation function of Appendix A.3 of NIST 800-38C and the formatting function as defined in Appendix A.2 of NIST 800-38C and `C` SHALL be a byte array containing the ciphertext as specified in Section 6.2 of NIST 800-38C. • If tag verification succeeds, the `success` output is `TRUE` and the `payload` array contains the decrypted payload `P`. • If tag verification fails, the `success` output is `FALSE` and the contents of the `payload` array is undefined.

3.7. Message privacy

Message privacy is implemented using a block cipher in CTR mode.

Mapping (Version 1.0)
Message privacy SHALL use the AES-CTR mode as defined in NIST 800-38A with the following parameters: • `int CRYPTO_PRIVACY_NONCE_LENGTH_BYTES := 13` • Key length SHALL be `CRYPTO_SYMMETRIC_KEY_LENGTH_BITS` bits.

3.7.1. Privacy encryption

Function and description

```
byte[lengthInBytes(M)]
Crypto_Privacy_Encrypt(
    SymmetricKey K,
    byte[lengthInBytes(M)] M,
    byte[CRYPTO_PRIVACY_NONCE_LENGTH_BYTES] N)
```

Performs the encryption of the message `M` using the key `K` and a nonce `N`; the output contains the data `M` encrypted.

